

Satwik Medipalli

Data Scientist / Data Analyst — Python · SQL · Machine Learning · BI

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SUMMARY

Data scientist (M.S. Business Analytics, UT Dallas 2026) with three analytics internships and four publicly deployed data products spanning causal inference on 5.6M+ pitches, survival modeling for hardware reliability, explainable ML, and multi-agent LLM systems. Every project below is live at a public URL.

DEPLOYED PROJECTS (BUILT DURING M.S., 2024 – 2026)

- ShapeShift: Causal Evaluation of MLB Pitch Redesigns** | *Python, DuckDB, scikit-learn* | [live site](#) Spring 2026
- Measured whether pitch-design changes actually help using 5.6M+ Statcast pitches in a reproducible DuckDB warehouse: outcome-blind changepoint detection with thresholds frozen on 2024 data, then a locked 2025 matched event study (381 matched pairs, not-yet-treated controls)
 - Reported the aggregate effect honestly as a null (+0.02 RV/100, CI includes zero), isolated a matched featured case at -0.29 RV/100, and translated the evidence into Keep / Reshape / De-emphasize recommendations
- SiliconSentinel: Hardware Fleet Reliability** | *Python, XGBoost, lifelines* | silicon-sentinel.vercel.app Fall 2025
- Modeled 30-day failure risk for a 10,000-device semiconductor fleet across 12 sites with survival analysis and XGBoost remaining-useful-life scoring over physics-grounded failure simulation (Arrhenius, Black, Peck, Coffin-Manson) — 83% recall at 80% precision
 - Surfaced predictions in an ops console with an RMA-vs-monitor decision queue and SHAP deep dives
- NFL Draft Intelligence** | *Python, XGBoost, SHAP, SQL* | nfl-draft-intelligence.vercel.app Spring 2025
- Trained position-specific XGBoost classifiers on two decades of NFL Combine data to evaluate draft prospects, with SHAP values explaining which measurables drive each prediction and a cosine-similarity engine matching prospects to historical comps
- Diligence AI: Automated Financial Due Diligence** | *LangGraph, Pinecone, RAG* | diligence-ai-nine.vercel.app Spring 2026
- Built a multi-agent AI platform where six agents read SEC 10-K filings, extract financials, classify risk factors, and assemble an analyst-style report in minutes, with a source-cited Q&A interface over every document

EXPERIENCE

AI Research Intern Jan 2024 · Apr 2024
Boltzmann Labs

- Wrote a format-aware Python preprocessing engine that auto-detects three types of incoming biological data across ~500K records at 85%+ accuracy and routes each through the correct pipeline, replacing a fully manual data-preparation workflow
- Deployed the ML backend with Docker, Kubernetes, and AWS S3 behind REST APIs, turning a local prototype into a scalable cloud service

Data Analytics Intern Aug 2023 · Dec 2023
Bostonlogix

- Migrated ~1M records from MongoDB to SQL Server with Python to create a queryable analytics store, and audited and restored roughly 1,000 failing API endpoints with Postman
- Built an NLP extraction pipeline with OpenAI APIs on Microsoft Azure that pulls structured fields from ~100K unstructured text records, fully automating a manual review process
- Designed KPI dashboards in Bold BI for Hungry Jack's surfacing the store-performance metrics leadership relied on in shareholder reviews

BI Developer Intern Apr 2023 · Jul 2023
Ideabytes

- Built a real-time ingestion pipeline that streamed ~500K JSON sensor datapoints through APIs into Power BI, and presented the live dashboard directly to the CEO
- Developed linear regression and ARIMA models forecasting sensor performance at 92–95% one-month accuracy and 80–90% at twelve months, enabling proactive maintenance scheduling

EDUCATION

University of Texas at Dallas Richardson, TX
M.S. Business Analytics May 2026

GITAM University Hyderabad, India
B.Tech Computer Science with Data Science May 2024

TECHNICAL SKILLS

Analysis & ML: Python (Pandas, NumPy, Scikit-learn), SQL, R, XGBoost, SHAP, causal inference (matched event studies), survival analysis (lifelines), forecasting (ARIMA, regression), statistics

BI & Visualization: Power BI, Tableau, Bold BI, Streamlit, Plotly, KPI design, executive reporting

Data Engineering: PostgreSQL, SQL Server, MongoDB, DuckDB, AWS S3, Microsoft Azure, FastAPI, Flask, Git, Postman

AI / LLM: OpenAI APIs, LangChain, LangGraph, RAG, Pinecone, vector search, LLM evaluation